

 Excel Ribbon & Grid Formulas & Calculations

INSTANT

 Data Analysis Diploma • Session 02

EXCEL BASICS

Interface, Formulas & Core Functions for Modern Data Analysts

6 Pillars

Curriculum Roadmap

Interface • Structure •
Formulas

29 Slides

Applied Core Modules

Real-world practice & labs

4 Core Groups

Function Tech Stack

Stats • Logic • Text • Math

[Start Session 02 →](#) Cell References (\$) Core Functions & Logic

— LEARNING OBJECTIVES

Session Objectives (6 Core Pillars)

By the end of this session, you will be able to master 6 core pillars of Excel:



Navigate Excel efficiently

Move confidently around the ribbon, sheets, and workbook.



Understand workbook structure

Know how workbooks, worksheets, and cells relate.



Create formulas correctly

Write clean, accurate formulas using proper syntax.



Master cell references

Apply relative, absolute, and mixed references with confidence.



Use core Excel functions

Apply the most common statistical, logical & text functions.



Perform basic data analysis

Summarize and interpret data like a real analyst.

OVERVIEW

What is Microsoft Excel?

A spreadsheet application for organizing, calculating, and analyzing data in rows and columns.



A spreadsheet application for organizing, calculating, and analyzing data in rows and columns.

WHY DATA ANALYSTS USE IT

- Fast, flexible calculations on any size dataset
- Built-in functions for statistics & logic
- Universal format every business understands
- Perfect for prototyping before scaling to code

REAL-WORLD USE CASES



Financial Analysis

Budgets, forecasts, and P&L modeling.



HR Reporting

Headcount, salary bands, attendance tracking.



Sales Performance

Pipeline tracking and commission calculations.



Operations

Inventory counts and process dashboards.

FILE ORGANIZATION

Workbook vs. Worksheet

Understanding how Excel structures data files and individual grid sheets.



Workbook Structure

The entire Excel file (.xlsx) — a container for one or more worksheets.



Sales

Worksheet

Sheet 1



Employees

Worksheet

Sheet 2



Summary

Worksheet

Sheet 3

FILE CONTAINER

Workbook

One saved Excel file (.xlsx) that can hold many worksheets — think of it as a binder.

SINGLE GRID

Worksheet

A single grid/tab inside the workbook where your data actually lives.

DATA TABS

Multiple Sheets

Use separate sheets to organize related data — e.g. one per month, region, or team.

GRID ELEMENTS

Rows, Columns & Cells

The fundamental building blocks of every Excel spreadsheet grid.

Click any cell to test selection:

C2

	A	B	C	D	E
1					
2			C2		
3					
4					
5					

Active Cell (C2)

Range (B2:C4)

Column

Vertical, labeled with letters (A, B, C...).

Row

Horizontal, labeled with numbers (1, 2, 3...).

Cell

The intersection of a row and a column.

Cell Address

Column letter + row number, e.g. C2.

Range

A group of cells, e.g. B2:C4.

Active Cell

The cell currently selected and ready for input.

CORE INPUT TYPES

Data Types in Excel

Understanding how Excel handles and formats different kinds of information.



Number

1250

Quantities, IDs, measurements.



Text

"Cairo"

Names, labels, categories.



Date

27/07/2026

Timelines, deadlines, logs.



Currency

\$4,500.00

Prices, salaries, budgets.



Percentage

18%

Growth rates, ratios, shares.



Boolean

TRUE / FALSE

Logical checks & flags.

— EFFICIENT WORKFLOW

Basic Navigation & Shortcuts

Essential keyboard shortcuts and view modes to navigate large datasets quickly.

Ctrl + Arrow

Jump to the edge of a data block.

Ctrl + Home

Jump straight to cell A1.

Ctrl + End

Jump to the last used cell.

Shift + Arrow

Extend a selection cell by cell.



Selecting Cells & Ranges

Click a cell to select it. Click-drag, or Shift + click, to select a range. Ctrl + click to select non-adjacent cells.



Freeze Panes

View → Freeze Panes keeps header rows or columns visible while you scroll through large datasets.

**Don't merge cells**

Merged cells break sorting, filtering, and formulas.

**Keep one header row**

A single clear header keeps tools like Tables and PivotTables reliable.

**Avoid blank rows**

Gaps break AutoFill, ranges, and Table detection.

**Convert to Tables (Ctrl+T)**

Tables auto-expand ranges and structure formulas.

**Keep consistent formatting**

Uniform dates/numbers prevent silent calculation errors.

— SYNTAX & RULES

Introduction to Formulas

A formula is an instruction that tells Excel to calculate a result from values or cell references.

=

A formula is an instruction that tells Excel to calculate a result from values or cell references.

=A1+A2

Every formula starts with an equals sign (=)

RULES OF A FORMULA

- Always begins with =
- Can combine numbers, cell references, and operators
- Functions use parentheses: =SUM(A1:A5)
- Press Enter to calculate, F2 to edit

FORMULA BAR

fx | =A1+A2

The Formula Bar shows exactly what's typed in the active cell — even if the cell itself displays only the calculated result.

— MATH SYMBOLS

Arithmetic Operators

The core mathematical symbols used for formulas in Excel.

					
Addition	Subtraction	Multiplication	Division	Exponent	Grouping
<code>=A1+B1</code>	<code>=A1-B1</code>	<code>=A1*B1</code>	<code>=A1/B1</code>	<code>=A1^2</code>	<code>=(A1+B1)*C1</code>

Practical example: to calculate total pay from hourly rate and hours worked, use `=B2*C2` (rate × hours).

— CALCULATION ORDER

Order of Operations (PEMDAS)

Excel follows the same math order you learned in school — PEMDAS.

P

Parentheses

E

Exponents

M

Multiplication

D

Division

A

Addition

S

Subtraction

`=(B2+B3)*2`

- 1. Parentheses first: $B2 + B3 = 400 + 600 = 1,000$
- 2. Then multiply: $1,000 \times 2 = 2,000$

— RELATIVE REFERENCE

Relative Reference

The default reference type. When copied, it shifts automatically to match its new position.

A	B	C
10	20	=A1+B1 → 30
15	25	=A2+B2 → 40
5	35	=A3+B3 → 40

Data Analysis scenario: calculating "Total = Price × Quantity" for every row of a sales sheet — one formula, copied down hundreds of rows.

WHY IT MATTERS FOR ANALYSTS

- Copying =A1+B1 down to row 2 becomes =A2+B2 automatically.
- Perfect for repeating the same calculation across many rows.
- Default type whenever you type a plain cell reference.

ABSOLUTE REFERENCE

Absolute Reference

Locked with \$ signs. Stays fixed on the same cell no matter where the formula is copied.



=B2*D\$5 (Mixed (Row Locked Only))

Press F4

Item	Price	Tax Rate	Tax
Laptop	20,000	-	=B2*D\$5
Mouse	450	-	=B3*D\$5
Monitor	3,200	-	=B4*D\$5
Rate →		14% (D5)	

WHY IT MATTERS

Every row needs to multiply by the same fixed tax rate in D5 — the \$D\$5 anchor keeps pointing to that one cell as the formula is copied down.

💡 Press F4 to quickly toggle a reference between relative, absolute, and mixed.

MIXED REFERENCE

Mixed Reference

Locks only the column OR the row, letting the other side shift freely when copied.

\$A1

Column locked, row shifts

Copying across keeps column A; copying down moves the row number.

A\$1

Row locked, column shifts

Copying across moves the column letter; copying down keeps row 1.

Data Analysis Scenario: Multiplication Table: To build a grid where every row multiplies by a row-header value and every column multiplies by a column-header value, use $=\$A2*B\1 — this is the classic case where BOTH a locked column (\$A2) and a locked row (B\$1) work together in the same formula.

COMPARISON TABLE

Reference Types at a Glance

Quick comparison of all cell reference types in Excel.

Type	Example	Column	Row	Best Used For
Relative	A1	Shifts	Shifts	Repeating the same calc down many rows
Absolute	\$A\$1	Locked	Locked	A fixed constant (tax rate, exchange rate)
Mixed (col)	\$A1	Locked	Shifts	Grids referencing one fixed column
Mixed (row)	A\$1	Shifts	Locked	Grids referencing one fixed row

🔑 Key Takeaway: press F4 while editing a formula to cycle through all four reference types instantly — no need to type \$ signs by hand.

— EFFICIENT WORKFLOW

Copying Formulas

Techniques for quickly propagating formulas across large columns and grids.



Fill Handle

Drag the small square at the bottom-right corner of a cell to copy a formula into adjacent cells.



AutoFill

Excel detects a pattern (numbers, dates, formulas) and continues it automatically as you drag.



Double-Click Fill Handle

Double-click the fill handle to instantly copy a formula down an entire adjacent data column.

Illustration: dragging C2 (=A2*B2) down to C5 (=A5*B5)

— TROUBLESHOOTING

Common Formula Errors

Identify and fix common calculation error codes in Excel.

#DIV/0!

Dividing by zero or an empty cell.

#VALUE!

Wrong data type used in a calculation.

#NAME?

Excel doesn't recognize a function/na...

#REF!

A referenced cell was deleted.

#DIV/0!

Dividing by zero or an empty cell.

Faulty Formula:

=A2/B2 when B2 is blank

✓ **Recommended Fix:** Wrap in IFERROR or check the denominator first.

Tip: Always inspect referenced data types & ranges

#ErrorDiagnostic



Use Parentheses

Group calculations explicitly instead of relying on default order of operations.



Avoid Hard-Coding

Reference a cell instead of typing a fixed number directly into a formula.



Reference Cells, Not Values

=B2*C2 updates automatically; =20*5 never does.



Keep Formulas Readable

Break long formulas into helper columns when they get hard to read.

CORE AGGREGATIONS

SUM, AVERAGE, MIN & MAX

Essential statistical functions to summarize numerical columns.



SUM

`=SUM(B2:B10)`

Adds all values in a range.

Use case: Total monthly sales.



AVERAGE

`=AVERAGE(B2:B10)`

Calculates the mean of a range.

Use case: Average order value.



MIN

`=MIN(B2:B10)`

Finds the smallest value.

Use case: Lowest price in stock.



MAX

`=MAX(B2:B10)`

Finds the largest value.

Use case: Top performer's sales.

DATA INSPECTION

COUNT, COUNTA & COUNTBLANK

Functions for counting dataset entries and checking data completeness.



COUNT

`=COUNT(B2:B10)`

Counts cells containing numbers only.



COUNTA

`=COUNTA(B2:B10)`

Counts all non-empty cells (any type).



COUNTBLANK

`=COUNTBLANK(B2:B10)`

Counts empty cells in a range.



Business Use Case — Data Quality Check: Compare COUNTA (total filled entries) against COUNT (numeric entries only) in a "Salary" column — a gap between the two numbers instantly flags rows where salary was entered as text or left blank, before it breaks a downstream SUM or AVERAGE.

— DECISION LOGIC

The IF Function

Tests a condition, then returns one value if TRUE and another if FALSE.

```
=IF(logical_test, value_if_true, value_if_false)
```

Cell B2 Value (Score):

75



Condition: B2 >= 60 → TRUE

TRUE → "Pass"

FALSE → "Fail"

BUSINESS EXAMPLES

Sales bonus:

```
=IF(C2>10000,"Bonus","No Bonus")
```

Attendance flag:

```
=IF(D2<90%,"At Risk","OK")
```

Stock alert:

```
=IF(E2<10,"Reorder","In Stock")
```

— ROUNDING RULES

ROUND, ROUNDUP & ROUNDDOWN

Functions for controlling decimal precision in calculated metrics.

Test input value (A2):

4.567

12.894

99.449

3.14159

ROUND

`=ROUND(A2, 2)`

Rounds to nearest value

4.567 → 4.57

ROUNDUP

`=ROUNDUP(A2, 2)`

Always rounds away from zero

4.567 → 4.57

ROUNDDOWN

`=ROUNDDOWN(A2, 2)`

Always rounds toward zero (truncates)

4.567 → 4.56

Real use case: rounding calculated prices to 2 decimal places (cents) before displaying them to customers, or rounding units shipped down (ROUNDDOWN) so you never promise more stock than truly available.

— CRITERIA FILTERING

SUMIF, COUNTIF & AVERAGEIF

Perform mathematical and statistical aggregations based on specific criteria.

\$

SUMIF

Finance

=SUMIF(C2:C10,"Sales",D2:D10)

Total expenses only in the "Marketing" or "Sales" department.

N

COUNTIF

HR

=COUNTIF(C2:C10,"Cairo")

Count employees based in the Cairo office.

%

AVERAGEIF

Sales

=AVERAGEIF(C2:C10,"Sales",E2:E10)

Average deal size for the Sales team only.

— STRING MANIPULATION

LEN, LEFT, RIGHT & MID

Extract specific characters or measure character lengths in text cells.

Try interactive text input (A2):

Data Analysis

T

LEN

=LEN(A2)

Result: "13"

Counts the number of characters in text.

T

LEFT

=LEFT(A2,3)

Result: "Data"

Extracts characters from the start.

T

RIGHT

=RIGHT(A2,2)

Result: "ysis"

Extracts characters from the end.

T

MID

=MID(A2,2,3)

Result: "ata "

Extracts characters from the middle.

— TEXT STANDARDIZATION

TRIM, UPPER, LOWER & PROPER

Clean and format messy text data before running calculations or joins.

Try interactive text input (A2):

Data Analysis



TRIM

=TRIM(A2)

Result: "Data Analysis"

Removes extra leading, trailing & double spaces.



UPPER

=UPPER(A2)

Result: "DATA ANALYSIS"

Converts text to all uppercase.



LOWER

=LOWER(A2)

Result: "data analysis"

Converts text to all lowercase.



PROPER

=PROPER(A2)

Result: "Data Analysis"

Capitalizes the first letter of each word.

SAMPLE DATASET

Meet the Employee Dataset

We'll use this sample dataset for every exercise in this section — follow along in your own workbook.

Employee ID	Name	Department	City	Salary	Bonus	Age
E001	Aya Hassan	Sales	Cairo	18,500	2,000	29
E002	Omar Nabil	Finance	Giza	22,000	1,500	34
E003	Sara Adel	HR	Cairo	15,750	1,000	26
E004	Karim Fathy	Sales	Alexandria	19,200	2,500	31

... continues for 40 rows in the practice workbook.

Fields: Employee ID • Name • Department • City • Salary • Bonus • Age

INTERACTIVE TASKS

Instructor-Led Exercises

7 step-by-step tasks applying formulas to the Employee Dataset.

1

Calculate totals

Find the total salary paid across all employees.

`=SUM(Salary range)`

2

Find averages

What is the average bonus given company-wide?

`=AVERAGE(Bonus range)`

3

Identify min & max

Find the youngest and oldest employee by Age.

`=MIN( ) / =MAX( )`

4

Count records

How many employees work in Sales?

`=COUNTIF(Dept,"Sales")`

5

Round salary values

Round every salary to the nearest 100.

`=ROUND(Salary,-2)`

6

Classify with IF

Flag employees as Senior or Junior by age.

`=IF(Age>=30,"Senior","Junior")`

7

Clean text

Standardize employee names for a clean report.

`=PROPER(TRIM(Name))`



Excel Interface

Ribbon, workbook & worksheet structure, navigation.



Rows, Columns & Cells

Addresses, ranges, and data types.



Formulas

Syntax, operators, and order of operations.



Cell References

Relative, absolute, and mixed references.



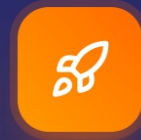
Core Functions

SUM, IF, ROUND, SUMIF, text functions & more.



Best Practices

Clean, error-free, analysis-ready spreadsheets.



Keep Practicing.

Every formula you write today builds the foundation for the data analysis skills ahead —
dashboards, PivotTables, and beyond.

Redo tonight's exercises from
memory

Explore 2–3 new formulas on
your own data

Bring questions to the next
session

Thank you — Instant Academy